

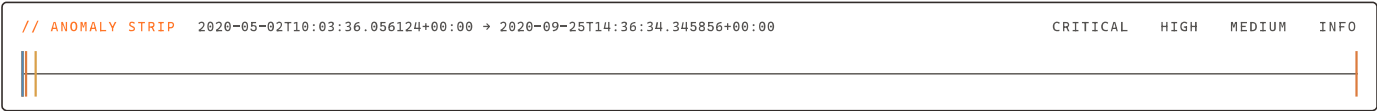
FORENSIC REPORT · USN \$J [323 020 RECORDS] [7 FINDINGS] [604800S BIN]

Activity timeline

critical

Automated parse of NTFS USN Journal \$J with behavioral analysis across 323 020 records. Most active path: ProgramData\Microsoft\RAC (78 781 events).

TOTAL RECORDS	CRITICAL	HIGH	MAX BURST	TIME SPAN
323 020 parsed	1 / 7 findings	4 findings	38 285 / 60s window	146 days



Findings

// BEHAVIORAL ANALYSIS

CRITICAL Timestamping signal: \$SI.Created earlier than \$FN.Created

category: timestamping_si_lt_fn · count: 6 650 · time: 2009-06-10T20:30:42.489868+00:00

6650 files where \$STANDARD_INFORMATION.Created is at least one second earlier than \$FILE_NAME.Created. \$FN times are normally write-protected by the kernel, so a back-dated \$SI is a classic forgery signal (Mimikatz timestamp, Cobalt Strike, SetFileTime). Investigate the listed files. Baseline filter excluded 60767/2794 (si/usec) routine entries.

\$ \$Recycle.Bin
\$ boot
\$ boot\cs-CZ
\$ boot\da-DK
\$ boot\de-DE
\$ boot\el-GR
\$ boot\en-US
\$ boot\es-ES
\$ boot\fi-FI
\$ boot\fonts

HIGH Timestamping signal: zeroed sub-second precision in \$SI

category: timestamping_usec_zeros · count: 447

447 files where at least one \$STANDARD_INFORMATION timestamp lands exactly on a 1-second boundary (microseconds == 0). NTFS preserves 100 ns resolution, so round timestamps usually come from explicit SetFileTime / API-driven manipulation. Cross-check with \$SI<\$FN finding for stronger signal. Baseline filter excluded 60767/2794 (si/usec) routine entries.

```
$ Windows\System32\MpSigStub.exe
$ Windows\Installer\27508a0.msp
$ Windows\System32\IEUDINIT.EXE
$ Windows\Installer\$\PatchCache$\Managed\E8EBC90469BFE03EA485673BA14799F\4.7.3062\microsoft.build.tasks.v4.0.dll_amd64
$ Windows\Installer\$\PatchCache$\Managed\E8EBC90469BFE03EA485673BA14799F\4.7.3062\microsoft.build.tasks.v4.0.dll_gac_x86
$ Windows\Installer\$\PatchCache$\Managed\E8EBC90469BFE03EA485673BA14799F\4.7.3062\microsoft.build.tasks.v4.0.dll_x86
$ Windows\Installer\$\PatchCache$\Managed\E8EBC90469BFE03EA485673BA14799F\4.7.3062\PenIMC_AMD64.dll
$ Windows\Installer\$\PatchCache$\Managed\E8EBC90469BFE03EA485673BA14799F\4.7.3062\PenIMC2_v0400_AMD64.dll
$ Windows\Installer\$\PatchCache$\Managed\E8EBC90469BFE03EA485673BA14799F\4.7.3062\PenIMC_v0400_AMD64.dll
$ Windows\Installer\$\PatchCache$\Managed\E8EBC90469BFE03EA485673BA14799F\4.7.3062\PenIMC_v0400_X86.dll
```

HIGH Executable files downloaded from the Internet (MOTW)

category: motw_executable_internet · count: 4

4 executable files (EXE/DLL/MSI/script/etc.) carry MOTW with ZoneId >= 3 (Internet/Restricted). Review the provenance URLs; correlate with execution evidence (Prefetch, Amcache, Shimcache) to confirm whether they ran.

```
$ Users\Misio1\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5\BLJE6BXH\Firefox Installer.exe | zone=3
$ Users\Misio1\Downloads\MinerGate-xFast-gui-1.4-win-amd.exe | zone=3 | host=https://download.minergate.com/xfast-win-gui-amd
$ Users\Misio1\Downloads\vc_redist.x64.exe | zone=3 | host=https://download.visualstudio.microsoft.com/download/pr/36c5faaf-bd8b-433f-b3d7-2af73bae10a8/212f41f2ccf6ee6d6dc27f901b7d77a1/vc_redist.x64.exe | ref=https://support.microsoft.com/pl-pl/help/2977003/the-latest-supported-visual-c-downloads
$ Users\Misio1\Downloads\tightvnc-2.8.11-gpl-setup-64bit.msi | zone=3 | host=https://www.tightvnc.com/download/2.8.11/tightvnc-2.8.11-gpl-setup-64bit.msi | ref=https://www.tightvnc.com/download.php
```

HIGH High-volume filesystem activity burst

category: burst_activity · count: 38 285 · time: 2020-09-25T11:22:55.566300+00:00



Observed 38285 USN records within 60 seconds. This may indicate automation, deployment, backup activity, malware activity, or ransomware-like behavior.

```
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\Z.Z...Z...ZZ.Z
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\ZZ.Z.Z..Z.ZZZ.ZZ
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\ZZZ.Z.....ZZ.Z
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\Z.Z.Z.Z.Z...ZZ.Z
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\Z.ZZZ...Z...Z
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\Z.ZZZZ.ZZ.Z.Z.Z
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\ZZ.Z.ZZ.ZZZZZZZZ
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\Z...Z..Z.ZZ...ZZ
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\ZZ.ZZ.ZZZ.ZZZZZZ
$ .\3590F75ABA9E485486C100C1A9D4FF062ZZZ..Z.....ZZZZZ\Z...ZZ.ZZZ..ZZ.Z
```

HIGH Mass rename activity detected

category: rename_wave · count: 1 511 · time: 2020-05-02T22:30:26.387128+00:00



Observed 1511 RENAME_NEW_NAME records. Mass renames can be normal in migrations, but are also a common ransomware signal.

```
$ Windows\AppCompat\Programs\AEINV_PREVIOUS.xml
$ Windows\AppCompat\Programs\AEINV_PREVIOUS.xml
$ 2273ae966936f9224e260bd541d4cbc6346d4af9
$ 2273ae966936f9224e260bd541d4cbc6346d4af9
$ Windows\SoftwareDistribution\Download\2273ae966936f9224e260bd541d4cbc6346d4af9
$ Windows\SoftwareDistribution\Download\2273ae966936f9224e260bd541d4cbc6346d4af9
$ Windows\AppCompat\Programs\AEINV_PREVIOUS.xml
$ Windows\AppCompat\Programs\AEINV_PREVIOUS.xml
$ Windows\AppCompat\Programs\AEINV_PREVIOUS.xml
$ Windows\AppCompat\Programs\AEINV_PREVIOUS.xml
```

MEDIUM Mass delete activity detected

category: delete_wave · count: 49 870 · time: 2020-05-02T10:05:36.748028+00:00



Observed 49870 FILE_DELETE records. This can indicate cleanup, user activity, software deployment, or destructive behavior.

```
$ ProgramData\Microsoft\RAC\Temp\sqlE82B.tmp
$ ProgramData\Microsoft\RAC\Temp\sqlE80B.tmp
$ ProgramData\Microsoft\RAC\Temp\sql3ED4.tmp
$ ProgramData\Microsoft\RAC\Temp\sql3EB4.tmp
$ ProgramData\Microsoft\RAC\Temp\sqlA1AC.tmp
$ ProgramData\Microsoft\RAC\Temp\sqlA17D.tmp
$ Windows\SoftwareDistribution\DataStore\Logs\tmp.edb
$ ProgramData\Microsoft\RAC\Temp\sqlB212.tmp
$ ProgramData\Microsoft\RAC\Temp\sqlB1F2.tmp
$ ProgramData\Microsoft\Windows Defender\Scans\History\Results\Quick\{0D48D493-1A1F-4A4F-A310-AC2888ADBEDD}
```

MEDIUM Potential tool or payload staging

category: suspicious_file_create · count: 245 · time: 2020-05-03T23:00:01.402450+00:00



Executable or script files were created in locations often used for staging.

```
$ Users\Administrator\AppData\Local\Temp\nnnqcdc-.dll
$ Users\Administrator\AppData\Local\Temp\nnnqcdc-.dll
$ Users\Administrator\AppData\Local\Temp\g-mwsa5t.dll
$ Users\Administrator\AppData\Local\Temp\g-mwsa5t.dll
$ Users\Administrator\AppData\Local\Temp\ywk6ymij.dll
$ Users\Administrator\AppData\Local\Temp\ywk6ymij.dll
$ Users\Administrator\AppData\Local\Temp\gbeuk895.dll
$ Users\Administrator\AppData\Local\Temp\gbeuk895.dll
$ Users\Administrator\AppData\Local\Temp\abfeljxg.dll
$ Users\Administrator\AppData\Local\Temp\abfeljxg.dll
```

INFO Alternate Data Streams observed

category: ads_present · count: 5

5 files carry at least one named \$DATA stream beyond the default. Most are benign (Zone.Identifier / SmartScreen / TXF metadata); see the separate finding for streams whose names are not on the known-benign list.

```
$ Users\Misio1\Downloads\MinerGate-xFast-cli-1.4-win-amd.zip:Zone.Identifier
$ Users\Misio1\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5\BLJE6BXH\Firefox Installer.exe:Zone.Identifier
$ Users\Misio1\Downloads\MinerGate-xFast-gui-1.4-win-amd.exe:Zone.Identifier
$ Users\Misio1\Downloads\vc_redist.x64.exe:Zone.Identifier
$ Users\Misio1\Downloads\tightvnc-2.8.11-gpl-setup-64bit.msi:Zone.Identifier
```

INFO Mark-of-the-Web (Zone.Identifier) on files

category: motw_observed · count: 5

5 files carry a populated Zone.Identifier ADS, marking them as downloaded from outside the local computer. ZoneId: 0=Local, 1=Intranet, 2=Trusted, 3=Internet, 4=Restricted. Inspect HostUrl/ReferrerUrl values for file provenance.

```
$ Users\Misio1\Downloads\MinerGate-xFast-cli-1.4-win-amd.zip | zone=3 | host=https://download.minergate.com/xfast-win-cli-amd
$ Users\Misio1\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5\BLJE6BXH\Firefox Installer.exe | zone=3
$ Users\Misio1\Downloads\MinerGate-xFast-gui-1.4-win-amd.exe | zone=3 | host=https://download.minergate.com/xfast-win-gui-amd
$ Users\Misio1\Downloads\vc_redist.x64.exe | zone=3 | host=https://download.visualstudio.microsoft.com/download/pr/36c5faaf-bd8b-433f-b3d7-2af73bae10a8/212f41f2ccf6e6d6dc27f901b7d77a1/vc_redist.x64.exe | ref=https://support.microsoft.com/pl-pl/help/2977003/the-latest-supported-visual-c-downloads
$ Users\Misio1\Downloads\tightvnc-2.8.11-gpl-setup-64bit.msi | zone=3 | host=https://www.tightvnc.com/download/2.8.11/tightvnc-2.8.11-gpl-setup-64bit.msi | ref=https://www.tightvnc.com/download.php
```

INFO

USN timeline summary

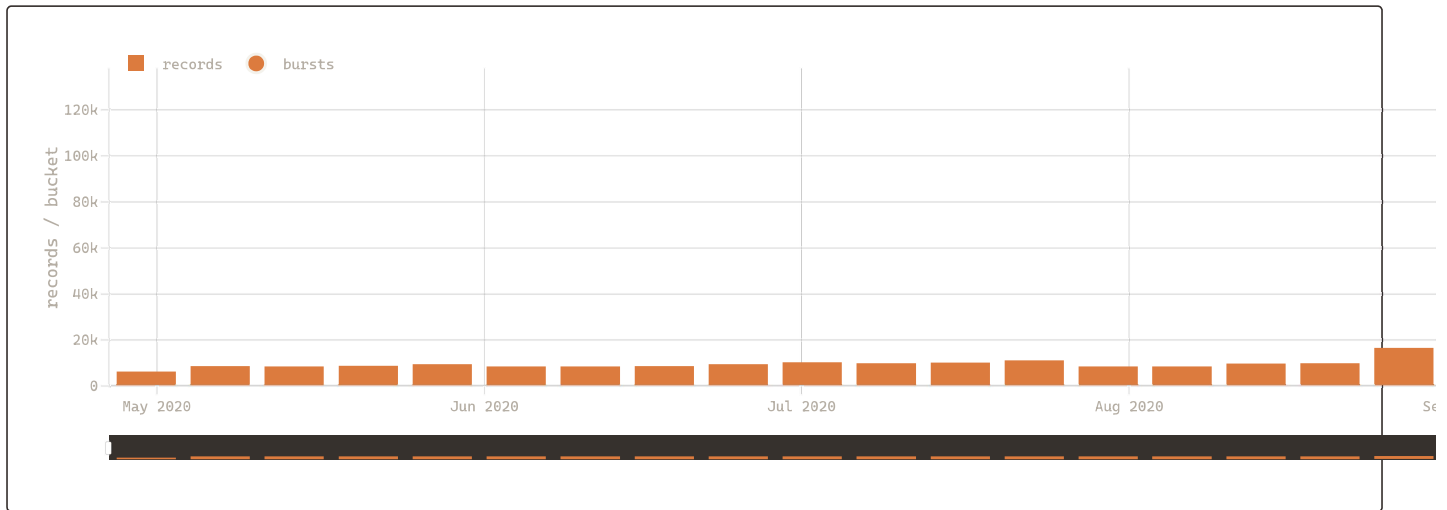
category: summary · count: 323 020 · time: 2020-05-02T10:03:36.056124+00:00


EVENTS ±15 MIN

Parsed 323020 records. Creates: 143098, writes: 202867, renames: 1511, deletes: 49870. Top reasons: [('CLOSE', 146251), ('FILE_CREATE', 143098), ('DATA_EXTEND', 138829), ('DATA_OVERWRITE', 72675), ('DATA_TRUNCATION', 50185)].

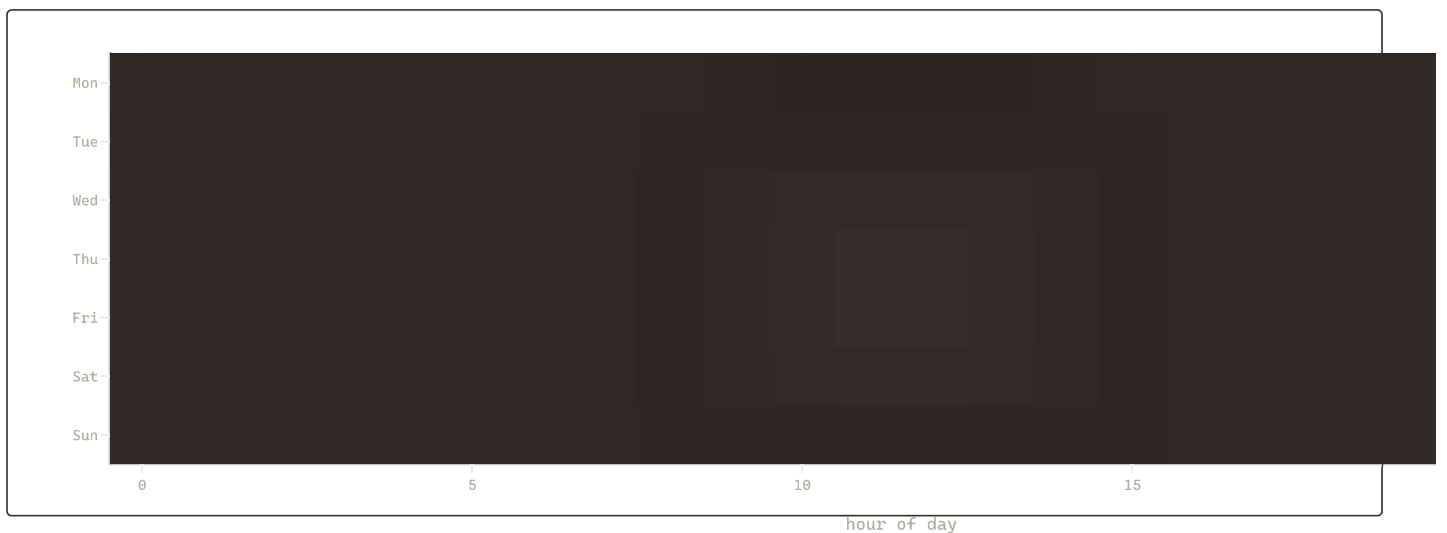
Activity timeline

// TIME-BUCKETED EVENT VOLUME



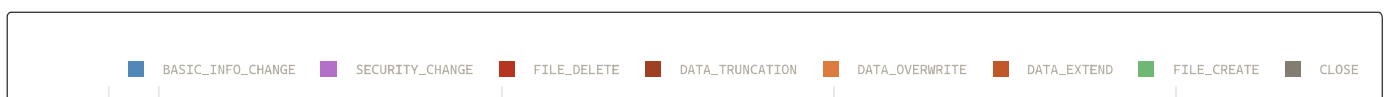
Hour x day heatmap

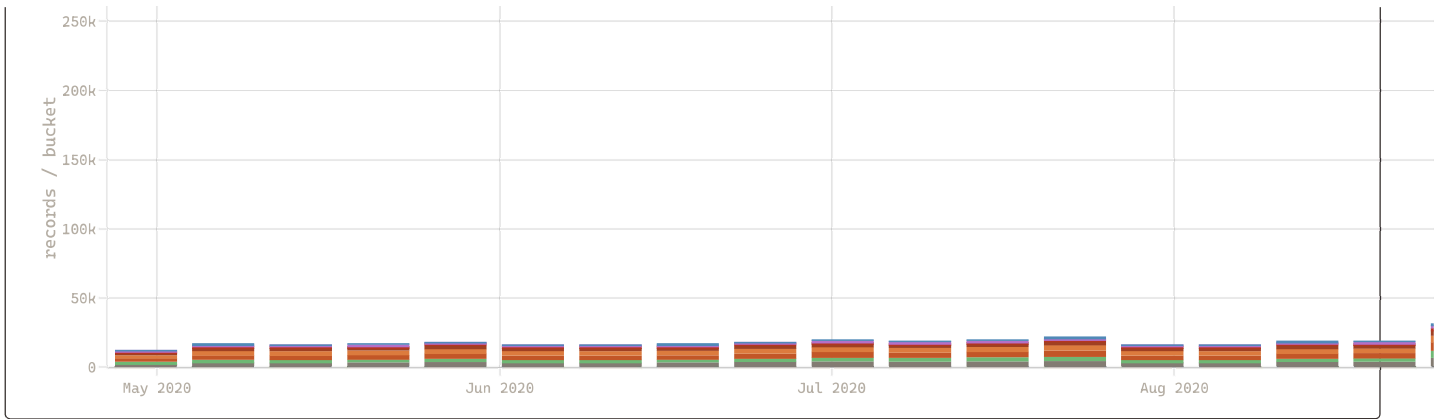
// OFF-HOURS ACTIVITY SURFACE



Reason flags over time

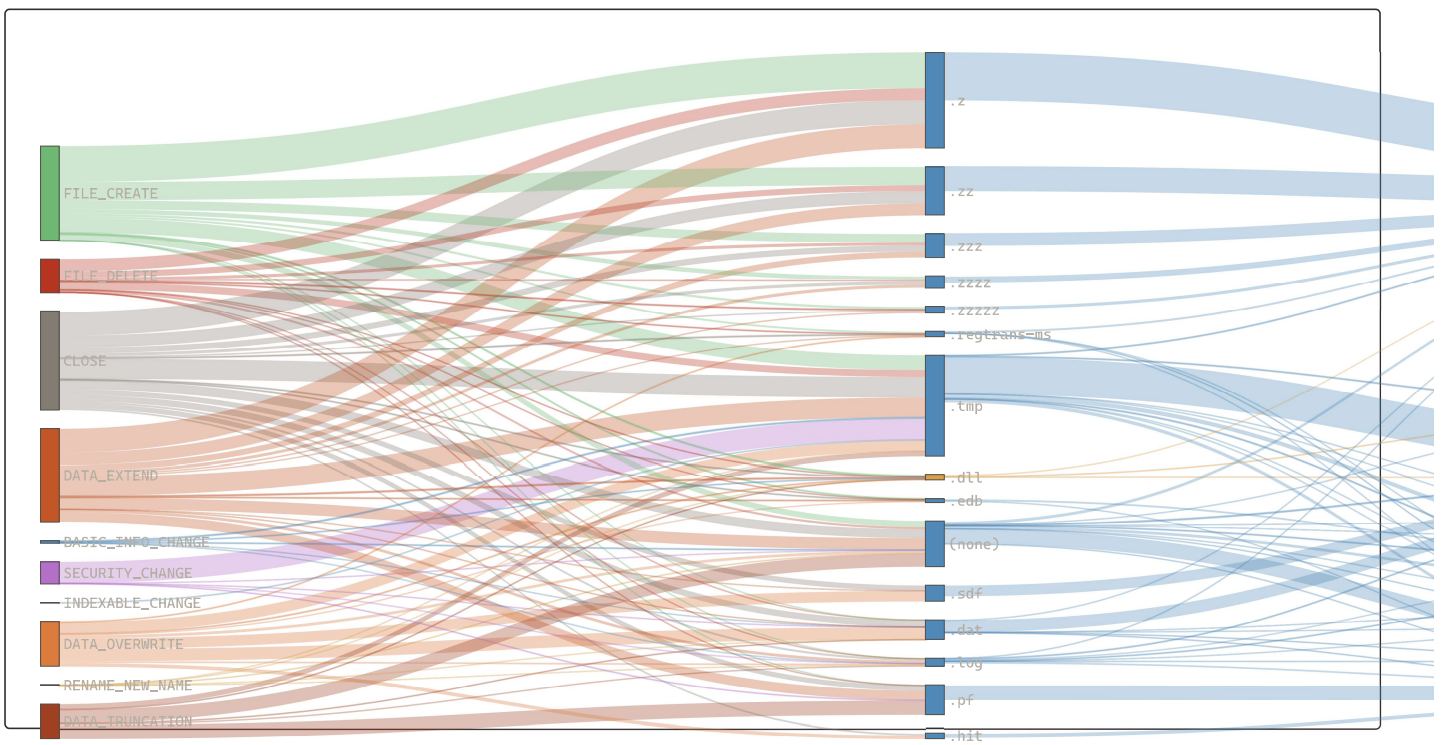
// STACKED EVENT TYPES





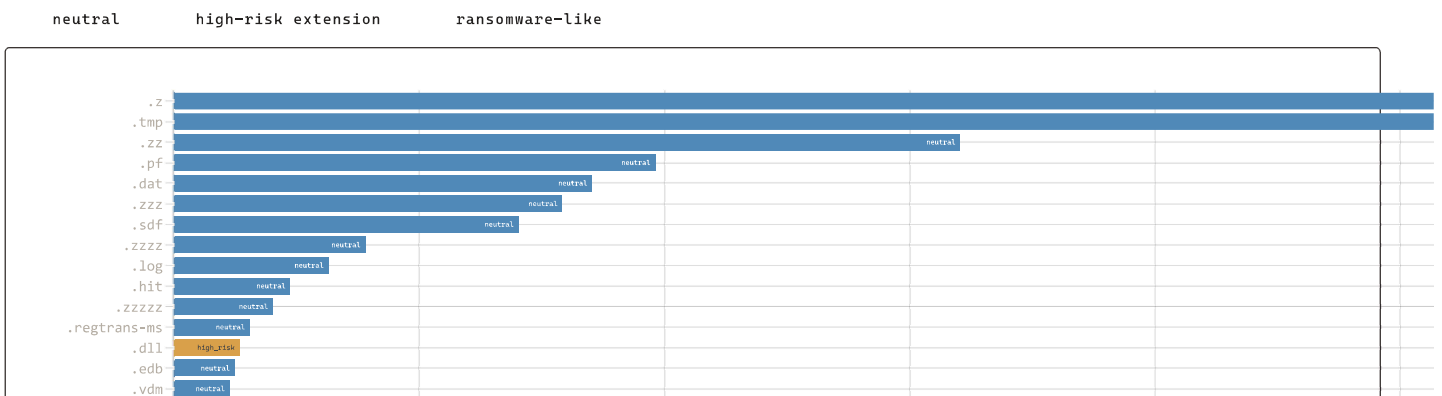
Reason flow

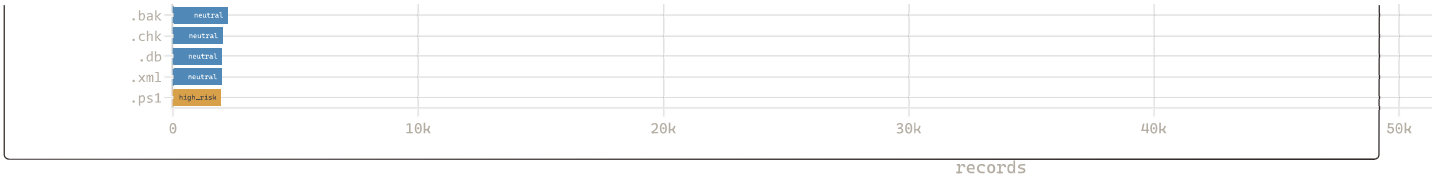
// REASON → EXTENSION → PATH BUCKET



Top file extensions

// RISK-CLASSED BY SIGNATURE LISTS





Path treemap

// DIRECTORY HOTSPOTS



Records

// FIRST 1 000 OF 323 020 ROWS · MULTI-FILTER · SORTABLE · VIRTUAL SCROLL

filter (/ to focus)...

CLEAREXPORT CSV1 000 / 1 000 (of 323 020)

REASON anyEXTENSION anyFROM YYYY-MM-DD HH:MMTO YYYY-MM-DD HH:MM

TIMESTAMP UTC ↕	REASON ↕	PATH ↕	FRN ↕	QUICK

ProgramData\Microsoft\Windows Defender



2020-05-02T10:05:36.6630...	DATA_OVERWRITE		ProgramData\Microsoft\RAC\I			
2020-05-02T10:05:36.6840...	DATA_OVERWRITE	CLOSE	tateData\RacWmiDataBookmark ProgramData\Microsoft\RAC\S	117430-2	F	P .dat
2020-05-02T10:05:36.6850...	DATA_OVERWRITE	CLOSE	s.dat tateData\RacWmiDataBookmark ProgramData\Microsoft\RAC\S	117430-2	F	P .dat
2020-05-02T10:05:36.6850...	DATA_OVERWRITE	CLOSE	s.dat tateData\RacWmiEventData\da ProgramData\Microsoft\RAC\P	117429-2	F	P .dat
2020-05-02T10:05:36.6850...	DATA_OVERWRITE		t ublishedData\RacWmiDatabas ProgramData\Microsoft\RAC\P	117398-3	F	P .sdf
2020-05-02T10:05:36.7470...	DATA_OVERWRITE	CLOSE	ublishedData\RacWmiDatabas ProgramData\Microsoft\RAC\P	117398-3	F	P .sdf
2020-05-02T10:05:36.7470...	DATA_OVERWRITE	DATA_EXTEND ...	ProgramData\Microsoft\RAC\I emp\sqlE82B.tmp	316-528	F	P .tmp
2020-05-02T10:05:36.7480...	FILE_DELETE	CLOSE	ProgramData\Microsoft\RAC\I emp\sqlE82B.tmp	316-528	F	P .tmp
2020-05-02T10:05:36.7480...	DATA_OVERWRITE		ProgramData\Microsoft\RAC\I tateData\RacDatabase.sdf	117397-3	F	P .sdf
2020-05-02T10:05:36.7480...	DATA_OVERWRITE	CLOSE	ProgramData\Microsoft\RAC\I tateData\RacDatabase.sdf	117397-3	F	P .sdf
2020-05-02T10:05:36.7490...	DATA_OVERWRITE	DATA_EXTEND ...	ProgramData\Microsoft\RAC\I emp\sqlE80B.tmp	299-448	F	P .tmp
2020-05-02T10:05:36.7490...	FILE_DELETE	CLOSE	ProgramData\Microsoft\RAC\I emp\sqlE80B.tmp	299-448	F	P .tmp
2020-05-02T10:05:36.7490...	DATA_OVERWRITE	CLOSE	ProgramData\Microsoft\RAC\I tateData\RacDatabase.sdf	32115-1	F	P .dat

Table is server-capped via --max-html-rows; the full event set is in CSV/JSON.